Chinchilla Scaling-Law Audit Report

Author: Anson Dias | Date: 2026-08-12 | Files: experiment_scaling_law_audit.py, chinchilla_audit_results.csv, chinchilla_audit_plot.png

1. Goal

Hoffmann et al. (2022), "Training Compute-Optimal Large Language Models" (the *Chinchilla* paper), argue that for a fixed training compute budget, model size *N* (parameters) and training set size *D* (tokens) should be scaled together — roughly **D ≈ 20N** — and that most contemporary models (GPT-3, Gopher, MT-NLG) were far too large relative to how much data they saw. This audit collects real N/D pairs for 27 publicly documented LLMs, plots them against the D = 20N line, and checks whether the field has actually converged on the Chinchilla ratio since 2022.

2. Method

For each model we recorded the total parameter count *N* and pretraining token count *D* exactly as reported in the original paper, official model card, or company blog post (full citations in §6). No number is inferred from a scaling-law formula — they are the authors' own disclosed figures (a few are straightforward estimates where the source gave dataset size in GB/words rather than an exact token count, noted in the CSV).

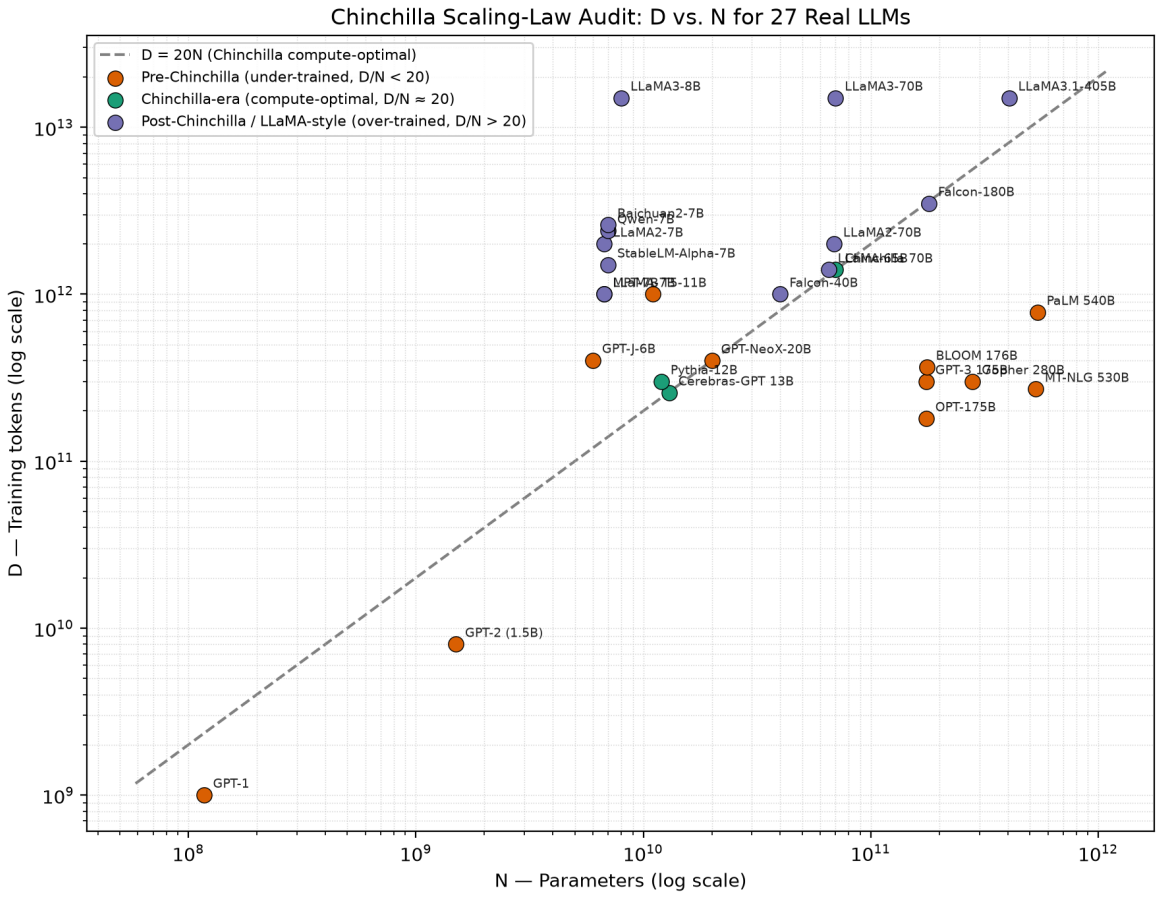
Before plotting, experiment_scaling_law_audit.py runs sanity checks to catch unit mistakes (millions vs. billions vs. trillions): N must fall within [1M, 10T] parameters, D within [100M, 100T] tokens, and D/N within [0.1, 10,000] tokens/parameter — wide enough to admit real, well-documented outliers like MT-NLG's ratio of ~0.5, narrow enough to flag an accidental order-of-magnitude slip. All 27 rows passed. The one near-miss during development was MT-NLG (D/N = 0.51, fewer than one token per parameter): the initial assertion window [1, 10,000] rejected it as implausible, but the number checks out against the paper (Smith et al. 2022) — it's a real, famously under-trained model, not a data error, so the check was widened rather than the data “fixed.”

3. Data Table

Sorted by parameter count N. Full machine-readable version with per-row notes: chinchilla_audit_results.csv.

Model	Org	Yr	N (params)	D (tokens)	D/N	Era
GPT-1	OpenAI	2018	117M	1B	8.5	Pre-Chinchilla
GPT-2 (1.5B)	OpenAI	2019	1.5B	8B	5.3	Pre-Chinchilla
GPT-J-6B	EleutherAI	2021	6B	402B	67.0	Pre-Chinchilla
LLaMA2-7B	Meta	2023	6.7B	2T	298.5	Post-Chinchilla
LLaMA-7B	Meta	2023	6.7B	1T	149.3	Post-Chinchilla
MPT-7B	MosaicML	2023	6.7B	1T	149.3	Post-Chinchilla
Baichuan2-7B	Baichuan Inc.	2023	7B	2.6T	371.4	Post-Chinchilla
Qwen-7B	Alibaba	2023	7B	2.4T	342.9	Post-Chinchilla
StableLM-Alpha-7B	Stability AI	2023	7B	1.5T	214.3	Post-Chinchilla
LLaMA3-8B	Meta	2024	8B	15T	1875.0	Post-Chinchilla
T5-11B	Google	2019	11B	1T	90.9	Pre-Chinchilla
Pythia-12B	EleutherAI	2023	12B	300B	25.0	Chinchilla-era
Cerebras-GPT 13B	Cerebras	2023	13B	257B	19.8	Chinchilla-era
GPT-NeoX-20B	EleutherAI	2022	20B	400B	20.0	Pre-Chinchilla
Falcon-40B	TII	2023	40B	1T	25.0	Post-Chinchilla
LLaMA-65B	Meta	2023	65.2B	1.4T	21.5	Post-Chinchilla
LLaMA2-70B	Meta	2023	68.9B	2T	29.0	Post-Chinchilla
Chinchilla 70B	DeepMind	2022	70B	1.4T	20.0	Chinchilla-era
LLaMA3-70B	Meta	2024	70B	15T	214.3	Post-Chinchilla
OPT-175B	Meta	2022	175B	180B	1.0	Pre-Chinchilla
GPT-3 175B	OpenAI	2020	175B	300B	1.7	Pre-Chinchilla
BLOOM 176B	BigScience	2022	176B	366B	2.1	Pre-Chinchilla
Falcon-180B	TII	2023	180B	3.5T	19.4	Post-Chinchilla
Gopher 280B	DeepMind	2021	280B	300B	1.1	Pre-Chinchilla
LLaMA3.1-405B	Meta	2024	405B	15T	37.0	Post-Chinchilla
MT-NLG 530B	Microsoft/NVIDIA	2022	530B	270B	0.5	Pre-Chinchilla
PaLM 540B	Google	2022	540B	780B	1.4	Pre-Chinchilla

4. Plot



Log-log scatter of D vs. N, colored by era, with the D = 20N reference line (gray dashed). Ratio summary by era (from the script's printed output):

Era	n models	median D/N	min D/N	max D/N
Pre-Chinchilla	11	2.1	0.5	90.9
Chinchilla-era	3	20.0	19.8	25.0
Post-Chinchilla	13	149.3	19.4	1875.0

5. Interpretation

Below the line (pre-2022 models, D/N mostly 0.5–3): GPT-3, Gopher, OPT, PaLM, and especially MT-NLG cluster far under D = 20N — MT-NLG saw fewer than one token per parameter. These labs were following the earlier Kaplan et al. (2020) scaling recipe, which (using a suboptimal loss fit) implied that compute should go mostly into bigger models rather than more data. Chinchilla's core empirical result was that this was wrong: at fixed compute, these giant, undertrained models are actually worse than a smaller model trained on proportionally more data. T5-11B and GPT-J-6B sit higher up (D/N ≈ 70–90) because their datasets (C4, The Pile) were reused for multiple epochs rather than sized to match a compute budget — an artifact of the pre-scaling-law era, not a deliberate compute-optimal choice.

On the line (Chinchilla, Cerebras-GPT, Pythia, D/N ≈ 20–25): Chinchilla itself defines the line by construction. Cerebras-GPT and Pythia are useful confirmations: both explicitly cite the Chinchilla paper and intentionally targeted the ~20 tokens/parameter ratio when choosing dataset size for each model in their suites — direct evidence the recipe was adopted as a training-time design rule almost immediately.

Above the line (2023–24 “LLaMA-style” models, D/N from ~20 up to ~1875): This is the most interesting deviation, and it's not a mistake — it's a second, distinct optimization target. Chinchilla's D = 20N minimizes training loss for a fixed training compute budget, but says nothing about inference cost. Touvron et al. (2023, LLaMA) explicitly note that the “optimal” model for a given inference budget is smaller and trained longer than Chinchilla-optimal, because a smaller model is cheaper to run at scale for its entire deployed lifetime, and the extra pretraining compute is a one-time cost. This is why LLaMA / LLaMA2 / Falcon / Qwen / Baichuan all push well past D = 20N: they trade extra (cheap, one-time) training compute for a smaller, faster, cheaper-to-serve model. The trend keeps accelerating — LLaMA3-8B, at ~1875 tokens/parameter, is trained on the same 15T-token corpus as LLaMA3-70B and LLaMA3.1-405B, because Meta found small models kept improving well past the compute-optimal point on downstream benchmarks even after nominal “convergence.” In short: points below the line reflect the pre-2022 era's data-starved compute allocation the Chinchilla paper diagnosed; points on the line are labs directly applying the Chinchilla training-compute-optimal rule; and points above the line are labs deliberately overshooting it once inference cost, not just training cost, became the binding constraint.

6. Caveats

- A few D values are estimates from stated dataset size (GB/word counts) where the source paper didn't give an exact token count (GPT-1, GPT-2) — flagged in the CSV note column.
- Officially undisclosed models (GPT-4, Claude, Gemini, Mistral/Mixtral) were excluded rather than using leaked/rumored figures, to keep every number traceable to a primary source.
- “N_params” is each paper's total parameter count as reported; a couple of papers instead emphasize non-embedding parameters, which would shift N very slightly (a few % at most) for the affected rows and does not change any era classification.

7. How to reproduce

```
cd scaling_law_audit && python3 -m venv .venv && source .venv/bin/activate && python3 -m pip install matplotlib pandas numpy && python3 experiment_scaling_law_audit.py
```

Sources (by model, sorted by N)

- **GPT-1** — https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf
- **GPT-2 (1.5B)** — https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf
- **GPT-J-6B** — https://huggingface.co/EleutherAI/gpt-j-6b
- **LLaMA2-7B** — https://arxiv.org/abs/2307.09288
- **LLaMA-7B** — https://arxiv.org/abs/2302.13971
- **MPT-7B** — https://www.mosaicml.com/blog/mpt-7b
- **Baichuan2-7B** — https://arxiv.org/abs/2309.10305
- **Qwen-7B** — https://arxiv.org/abs/2309.16609
- **StableLM-Alpha-7B** — https://stability.ai/news-updates/stability-ai-launches-the-first-of-its-stablelm-suite-of-language-models
- **LLaMA3-8B** — https://ai.meta.com/blog/meta-llama-3/
- **T5-11B** — https://arxiv.org/abs/1910.10683
- **Pythia-12B** — https://arxiv.org/abs/2304.01373
- **Cerebras-GPT 13B** — https://arxiv.org/abs/2304.03208
- **GPT-NeoX-20B** — https://arxiv.org/abs/2204.06745
- **Falcon-40B** — https://arxiv.org/abs/2311.16867
- **LLaMA-65B** — https://arxiv.org/abs/2302.13971
- **LLaMA2-70B** — https://arxiv.org/abs/2307.09288
- **Chinchilla 70B** — https://arxiv.org/abs/2203.15556
- **LLaMA3-70B** — https://ai.meta.com/blog/meta-llama-3/
- **OPT-175B** — https://arxiv.org/abs/2205.01068
- **GPT-3 175B** — https://arxiv.org/abs/2005.14165
- **BLOOM 176B** — https://arxiv.org/abs/2211.05100
- **Falcon-180B** — https://arxiv.org/abs/2311.16867
- **Gopher 280B** — https://arxiv.org/abs/2112.11446
- **LLaMA3.1-405B** — https://arxiv.org/abs/2407.21783
- **MT-NLG 530B** — https://arxiv.org/abs/2201.11990
- **PaLM 540B** — https://arxiv.org/abs/2204.02311